Data & Analytics Accelerator: Imgest-Mesh

Overview of Application for Industrial Inspection

Douglas Sayles, Trace3 – Data & Analytics douglas.sayles@trace3.com

January 31, 2026

Contents

1 Weather Prediction Markets Operating Book

A combined textbook + lab manual for learning, calibration, and disciplined profitability

---

1.1 Chapter 1: Purpose, Philosophy, and Learning Objective

This operating book exists to support learning-first participation in weather-based prediction markets, specifically temperature and precipitation markets. Profit is treated as validation of correct probabilistic thinking, not the primary goal.

The objective is to develop: - Accurate probability intuition - Discipline under uncertainty - Clear separation between signal and noise - Long-term survivability

Prediction markets are trades on mispriced probabilities, not bets on outcomes.

1.2 Chapter 2: Mental Model – Probability, Not Outcomes

Each contract pays 1 if an event occurs and 0 if it does not. Market price equals implied probability.

The question is never "Will this happen?" but: "Is the market probability wrong, and if so, by how much?"

1.3 Chapter 3: Market Selection – Scope and Constraints

This playbook applies only to: - Single-day temperature markets - Single-day precipitation markets

Geographic focus: - Phoenix metro (PHX settlement stations)

Excluded: - Multi-day aggregates - Narrative weather events - Extreme precipitation tails (≥1.00″) - Laddering or multi-leg trades

---

1.4 Chapter 4: Strategy Framework Overview

Primary Strategy: Threshold Fade
Always-On Lens: Station Awareness

Deferred (Phase 2+): - Late confidence fades - Seasonal base-rate overlays - Binary laddering

---

1.5 Chapter 5: Threshold Fade Strategy

Markets misprice probabilities near thresholds when the forecast mean approximates the cutoff and variance is wide.

Rules: - Trade only when market probability is 65–80% - Threshold must lie near forecast mean - Fade overstated confidence

---

1.6 Chapter 6: Temperature Markets

Characteristics: - Tight distributions - Lower variance - Strong station bias effects

Edge sources: - Overconfidence near round numbers - Ignoring wind, clouds, inversions

Execution: - Trade daily highs and lows - Fade confidence near median forecasts

---

1.7 Chapter 7: Precipitation Markets

Characteristics: - Skewed distributions - Fat tails - Localization sensitivity

Execution: - Focus on ≥0.10″, ≥0.25″, ≥0.50″ - Smaller position sizes - Avoid ≥1.00″ early

---

1.8 Chapter 8: Station Awareness

Settlement is mechanical, not experiential.

Rules: - Always verify settlement station - Nearby weather is irrelevant if it misses the sensor

---

1.9 Chapter 9: Bankroll Philosophy

Bankroll is learning capital.

Rules: - Fixed bankroll during Phase 1 - No top-ups or withdrawals - No confidence-based resizing

---

1.10 Chapter 10: Position Sizing

- Standard: 1% per trade
- Max: 2% after 20+ trades
- Max 3% per weather system

Caps: - Temperature: 1–1.5% - Precipitation: ≤1%

1.11 Chapter 11: Correlation Risk

Treat as correlated: - Same day - Same station - Same system

Never stack unknowingly.

1.12 Chapter 12: Drawdown Rules

Soft stop: - 10% drawdown □ cut size 50%, temp only

Hard stop: - 20% drawdown □ halt and review

1.13 Chapter 13: Trade Logging

Every trade must record: 1. Market & threshold 2. Market probability 3. Personal probability 4. Rationale 5. Invalidating conditions

---

1.14 Chapter 14: Research – Market Side

Study: - Price movement over time - Reaction to forecast updates - Liquidity changes - Narrative drift

---

1.15 Chapter 15: Research – Independent Analysis

Focus on: - Distributions, not point forecasts - Variance and confidence bands - Base rates and seasonality - Process failure points

---

1.16 Chapter 16: Reconciling Market vs Research

Quantify divergence. Identify assumptions. Trade only with explicit rationale.

1.17 Chapter 17: Phase Progression

Phase 1: Threshold Fade, Phoenix, calibration-first
Phase 2: Confidence fades, base-rate overlays
Phase 3: Laddering and structure analysis

1.18 Chapter 18: Operating Principles

- Boring is good
- Consistency beats cleverness
- Survival enables compounding
- Probability discipline is edge

This is a living document.